

**МИНИСТЕРСТВО ОБРАЗОВАНИЯ
РЕСПУБЛИКИ БЕЛАРУСЬ**
БЕЛОРУССКИЙ ГОСУДАРСТВЕННЫЙ УНИВЕРСИТЕТ
Факультет прикладной математики и информатики

КУРЕЦ ЛЮБОВЬ ОЛЕГОВНА

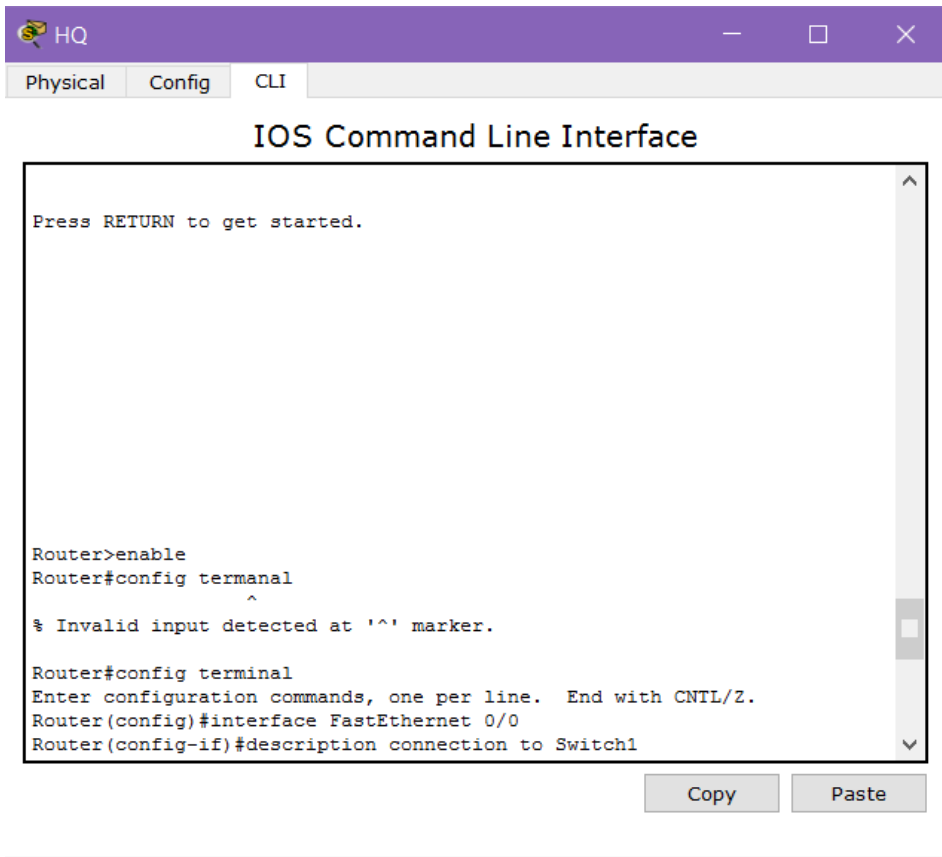
Отчет по лабораторной работе № 2 ,
вариант 11
студентки 3-го курса 2 -ой группы

Преподаватель
Рафеенко Е.Д.

2021 г.

Конфигурирование сетевых устройств

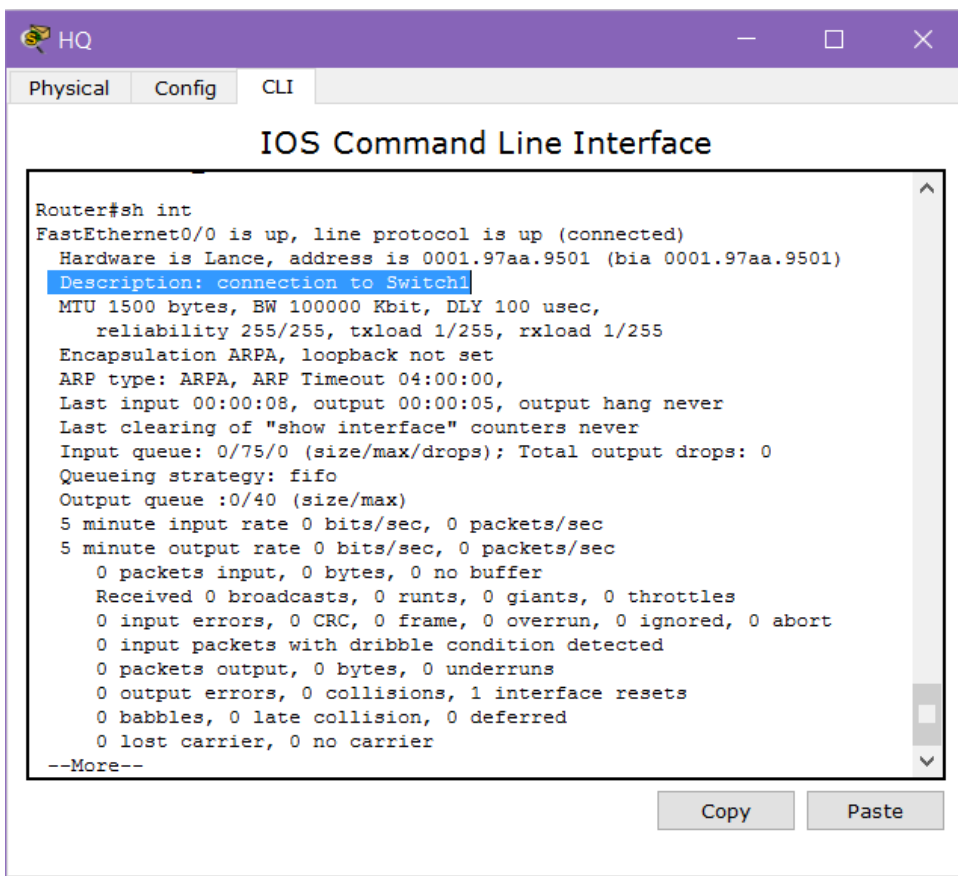
Задание 1 Добавление описания description *описание*



The screenshot shows the Cisco IOS Command Line Interface (CLI) window. The title bar indicates the user is logged in as 'HQ'. The tabs at the top are 'Physical', 'Config', and 'CLI'. The main window title is 'IOS Command Line Interface'. The terminal text shows the following sequence of commands and responses:

```
Press RETURN to get started.  
  
Router>enable  
Router#config terminal  
^  
% Invalid input detected at '^' marker.  
  
Router#config terminal  
Enter configuration commands, one per line. End with CNTL/Z.  
Router(config)#interface FastEthernet 0/0  
Router(config-if)#description connection to Switch1
```

At the bottom of the terminal window, there are 'Copy' and 'Paste' buttons.

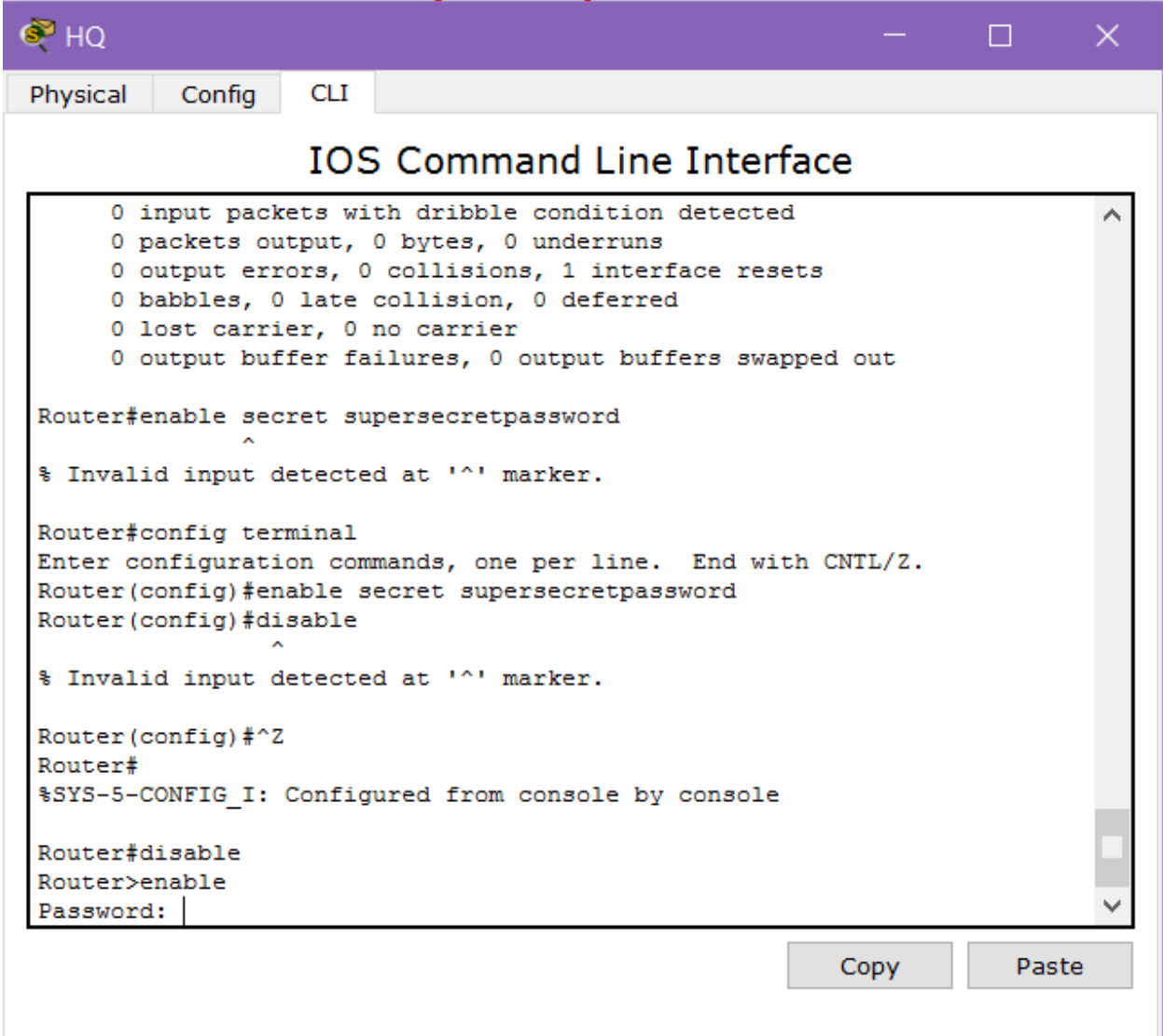


The screenshot shows the Cisco IOS Command Line Interface (CLI) window. The title bar indicates the user is logged in as 'HQ'. The tabs at the top are 'Physical', 'Config', and 'CLI'. The main window title is 'IOS Command Line Interface'. The terminal text shows the following sequence of commands and responses:

```
Router#sh int  
FastEthernet0/0 is up, line protocol is up (connected)  
  Hardware is Lance, address is 0001.97aa.9501 (bia 0001.97aa.9501)  
  Description: connection to Switch1  
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,  
    reliability 255/255, txload 1/255, rxload 1/255  
  Encapsulation ARPA, loopback not set  
  ARP type: ARPA, ARP Timeout 04:00:00,  
  Last input 00:00:08, output 00:00:05, output hang never  
  Last clearing of "show interface" counters never  
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0  
  Queueing strategy: fifo  
  Output queue :0/40 (size/max)  
  5 minute input rate 0 bits/sec, 0 packets/sec  
  5 minute output rate 0 bits/sec, 0 packets/sec  
    0 packets input, 0 bytes, 0 no buffer  
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles  
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort  
    0 input packets with dribble condition detected  
    0 packets output, 0 bytes, 0 underruns  
    0 output errors, 0 collisions, 1 interface resets  
    0 babbles, 0 late collision, 0 deferred  
    0 lost carrier, 0 no carrier  
  --More--
```

At the bottom of the terminal window, there are 'Copy' and 'Paste' buttons.

Задание 2 Установка секретного пароля enable secret *пароль*



The screenshot shows a network simulator window titled 'HQ' with tabs for 'Physical', 'Config', and 'CLI'. The 'CLI' tab is active, displaying the 'IOS Command Line Interface'. The terminal output shows the following sequence of commands and responses:

```
0 input packets with dribble condition detected
0 packets output, 0 bytes, 0 underruns
0 output errors, 0 collisions, 1 interface resets
0 babbles, 0 late collision, 0 deferred
0 lost carrier, 0 no carrier
0 output buffer failures, 0 output buffers swapped out

Router#enable secret supersecretpassword
^
% Invalid input detected at '^' marker.

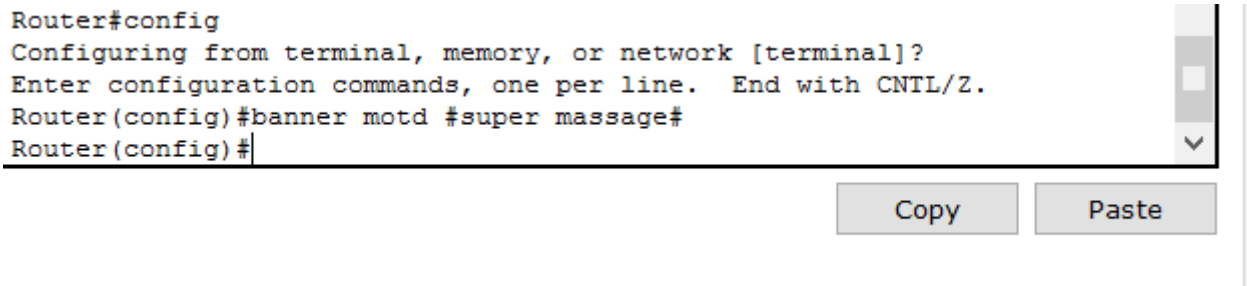
Router#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#enable secret supersecretpassword
Router(config)#disable
^
% Invalid input detected at '^' marker.

Router(config)#^Z
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#disable
Router>enable
Password: |
```

At the bottom of the CLI window, there are 'Copy' and 'Paste' buttons.

Задание 3 banner motd # message#



The screenshot shows the 'HQ' CLI interface with the 'CLI' tab active. The terminal output shows the following sequence of commands and responses:

```
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#banner motd #super message#
Router(config)#
```

At the bottom of the CLI window, there are 'Copy' and 'Paste' buttons.

Задание 4 Изменение имени роутера hostname *имя*

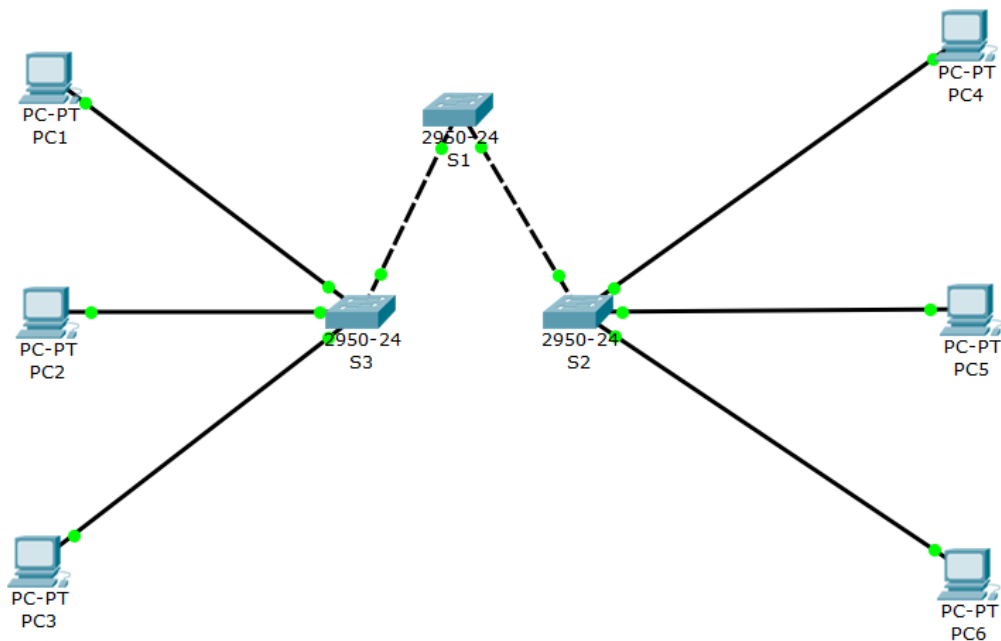

```
interface FastEthernet0/0
  description connection to Switch1
  no ip address
  duplex auto
  speed auto
!
interface FastEthernet0/1
  no ip address
  duplex auto
  speed auto
!
interface Serial0/1/0
  no ip address
!
interface Serial0/1/1
  no ip address
!
interface Vlan1
  no ip address
  shutdown
!
ip classless
!
ip flow-export version 9
!
!
!
banner motd ^Csuper massage^C
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
  login
!
!
!
end
```

FirstRouter#

Copy

Paste

Проектирование VLAN



Задание 1

Проверяем достижимость узлов в своих подсетях из PC1:

```
PC>ipconfig /all

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Physical Address.....: 0000.0CD3.A902
    Link-local IPv6 Address.....: ::
    IP Address.....: 172.17.10.21
    Subnet Mask.....: 255.255.0.0
    Default Gateway.....: 172.17.10.1
    DNS Servers.....: 0.0.0.0
    DHCP Servers.....: 0.0.0.0
    DHCPv6 Client DUID.....: 00-01-00-01-B8-C0-74-3C-00-00-0C-D3-A9-02
```

```
PC>ping 172.17.30.26

Pinging 172.17.30.26 with 32 bytes of data:

Reply from 172.17.30.26: bytes=32 time=1ms TTL=128
Reply from 172.17.30.26: bytes=32 time=0ms TTL=128
Reply from 172.17.30.26: bytes=32 time=0ms TTL=128
Reply from 172.17.30.26: bytes=32 time=0ms TTL=128

Ping statistics for 172.17.30.26:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Задание 2 Создаём VLANs на коммутаторе S1

```

S1(config)#vlan 10
S1(config-vlan)#name
% Incomplete command.
S1(config-vlan)#name Faculty/Staff
S1(config-vlan)#vlan 20
S1(config-vlan)#name Students
S1(config-vlan)#vlan 30
S1(config-vlan)#name Guest (Default)
S1(config-vlan)#vlan 99
S1(config-vlan)#name Management&Native
S1(config-vlan)#end
S1#

```

Аналогично на остальных коммутаторах

Задание 4

Проверка конфигурации VLANs на всех коммутаторах:

#show vlan brief

```
S2#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5, Fa0/7, Fa0/8, Fa0/9, Fa0/10, Fa0/12, Fa0/13, Fa0/14, Fa0/15, Fa0/16, Fa0/17, Fa0/19, Fa0/20, Fa0/21, Fa0/22, Fa0/23, Fa0/24, Gig0/1, Gig0/2
10	Faculty/Staff	active	Fa0/11
20	Students	active	Fa0/18
30	Guest (Default)	active	Fa0/6
99	Management&Native	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

S1#show vlan brief

VLAN Name	Status	Ports
1 default	active	Fa0/2, Fa0/4, Fa0/5, Fa0/6 Fa0/10 Fa0/14 Fa0/18 Fa0/22 Gig0/2
10 Faculty/Staff	active	
20 Students	active	
30 Guest (Default)	active	
99 Management&Native	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

S3#show vlan brief

VLAN Name	Status	Ports
1 default	active	Fa0/1, Fa0/2, Fa0/4, Fa0/5 Fa0/10 Fa0/15 Fa0/20 Fa0/24 Gig0/1, Gig0/2
10 Faculty/Staff	active	Fa0/11
20 Students	active	Fa0/18
30 Guest (Default)	active	Fa0/6
99 Management&Native	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

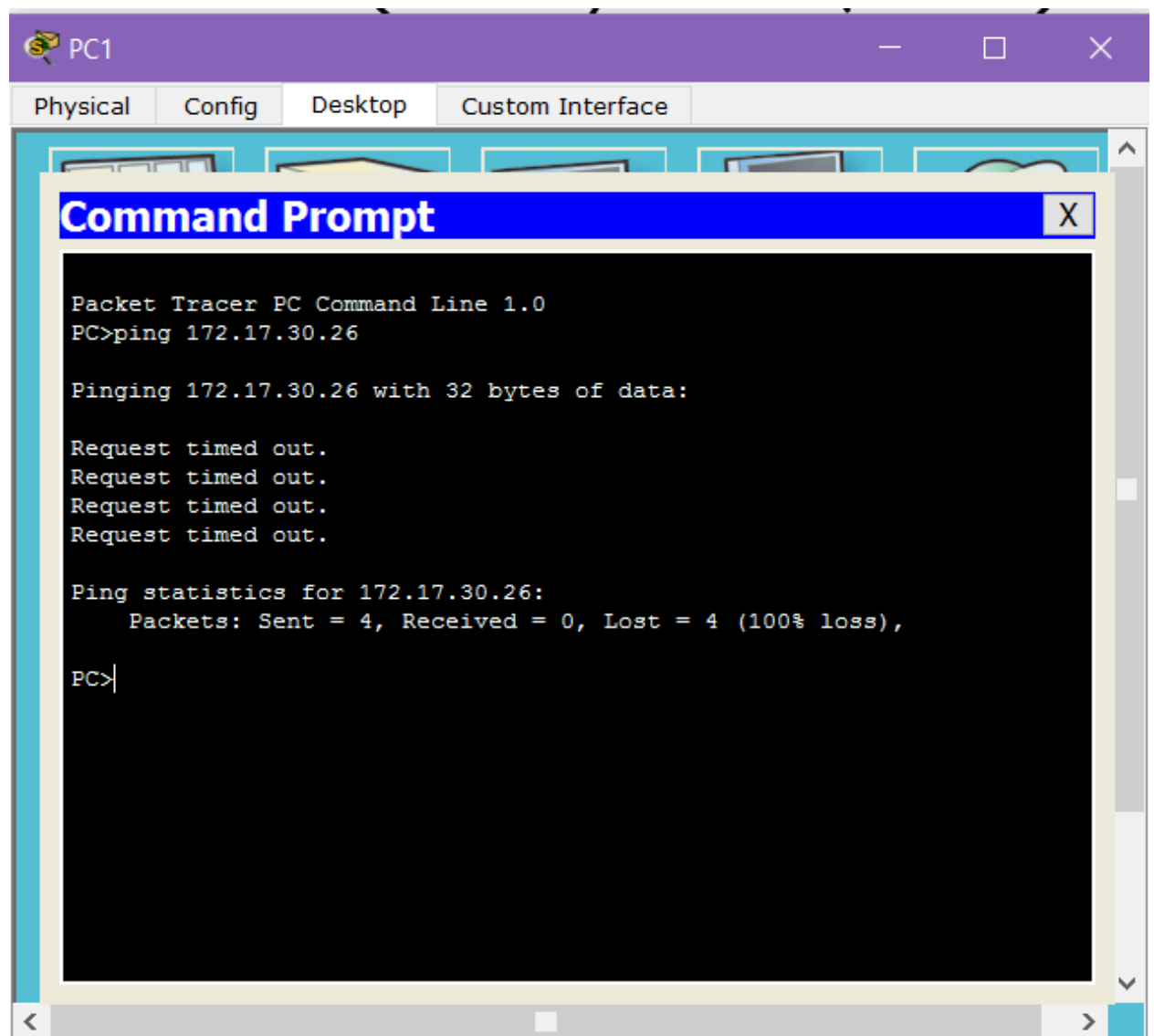
Задание 5

Назначение VLANs на порты:

```
S2#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
S2(config)#interface fastEthernet 0/6
S2(config-if)#switchport mode access
S2(config-if)#switchport access vlan 30
S2(config-if)#interface fastEthernet 0/11
S2(config-if)#switchport mode access
S2(config-if)#switchport access vlan 10
S2(config-if)#interface fastEthernet 0/18
S2(config-if)#switchport mode access
S2(config-if)#switchport access vlan 20
S2(config-if)#
```

Задание 6

Так происходит, потому что VLANs не настроены на S1



Задание 7

Сконфигурировать в S1 Fa0/1 и Fa0/3 как trunk и использование VLAN 99 как native VLAN:

```
S1(config)#interface FastEthernet0/1
S1(config-if)#switchport mode trunk

S1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state
down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state
up

S1(config-if)#switchport trunk native vlan 99
S1(config-if)#interface FastEthernet0/3
S1(config-if)#switchport mode trunk

S1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state
down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state
up

S1(config-if)#switchport trunk native vlan 99
```

Задание 8 Протестировать сеть. Использование команды *show vlan*

```
S1#show vlan
```

VLAN	Name	Status	Ports
1	default	active	Fa0/2, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1, Gig0/2
10	Faculty/Staff	active	
20	Students	active	
30	Guest(Default)	active	
99	Management&Native	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0
10	enet	100010	1500	-	-	-	-	-	0	0
20	enet	100020	1500	-	-	-	-	-	0	0
30	enet	100030	1500	-	-	-	-	-	0	0
99	enet	100099	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	0	0
1003	tr	101003	1500	-	-	-	-	-	0	0
1004	fdnet	101004	1500	-	-	-	ieee	-	0	0
1005	trnet	101005	1500	-	-	-	ibm	-	0	0

```
Remote SPAN VLANs
```

```
-----
Primary Secondary Type          Ports
-----
```

